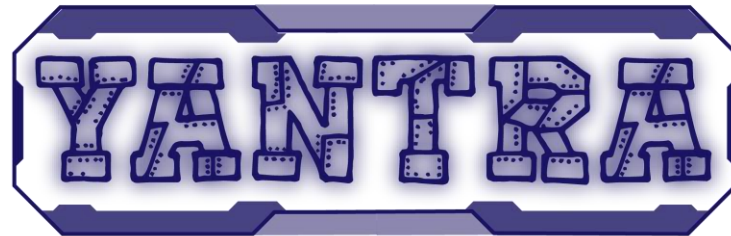


Laser engineered net shaping



- Laser Engineered Net Shaping (LENS), is perhaps **the first "true" direct-metal** rapid prototyping (RP) system, in that parts are full strength metals
- LENS process uses metal powders, per the user's preference, as build materials.
- The LENS 750 (12" x 12" x12") and LENS 850 (18" X 18" X 42") systems are manufactured and sold by **Optomec Design Company** in Albuquerque, NM.

Principle

- A high powered **Nd:YAG laser** focused onto a metal substrate creates a molten puddle on the substrate surface. **Powder** is then injected into the molten puddle to increase material volume.
- A “printing” motion system moves a platform horizontally and laterally as the laser beam traces the cross-section of the part being produced.



Meaning

Substrate

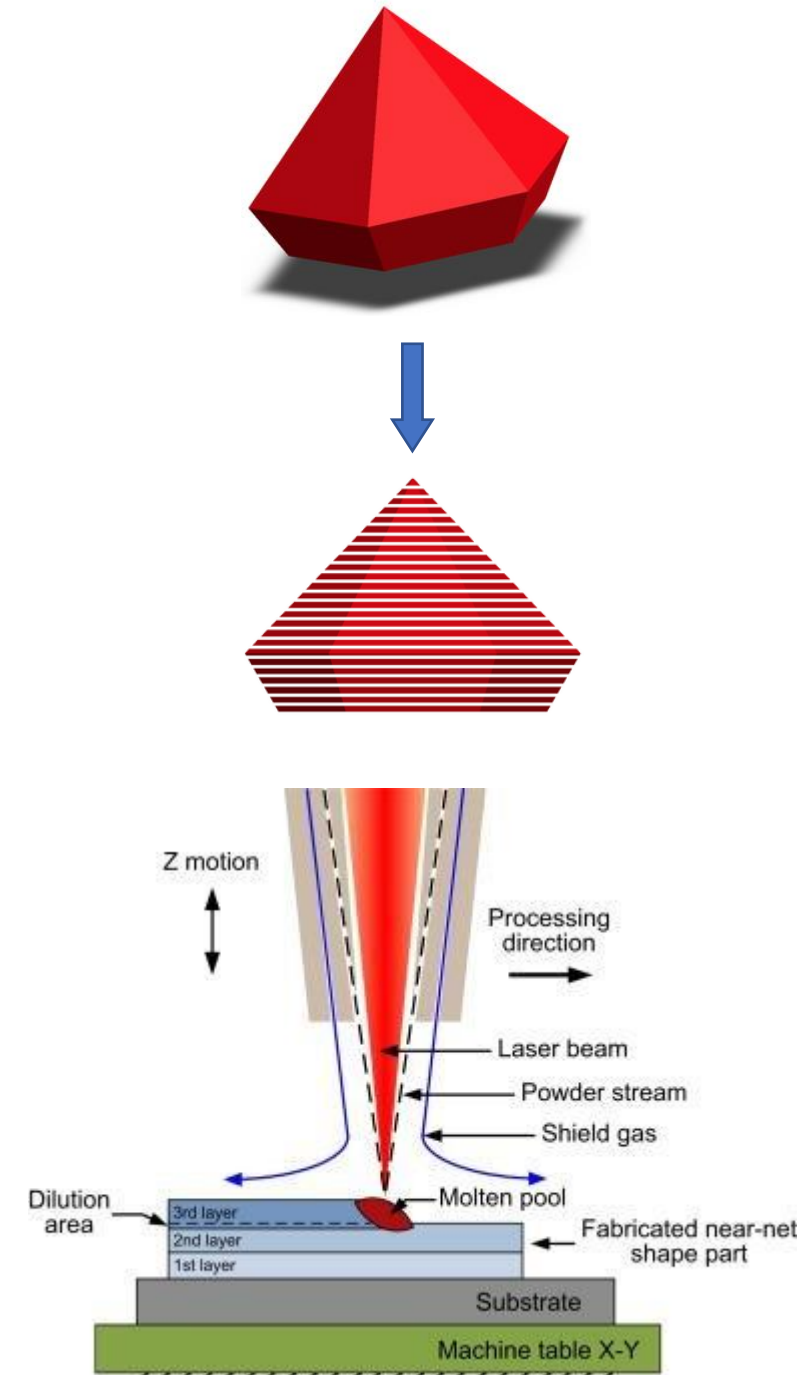
an underlying
substance or layer

PROCESS

Like most RP techniques, the LENS system uses a layered approach to manufacturing components, in which an **STL file** is sliced into horizontal cross sections, which are then downloaded to the machine from the bottom slice upwards..

A deposition head supplies metal powder to the focus of a high powered **Nd:YAG laser** beam to be melted.

This laser is typically directed by fiber optics or precision angled mirrors.



The fabrication process takes place in a low-pressure argon chamber for oxygen-free operation in the melting zone, ensuring that good adhesion is accomplished.

Also to isolate the process from the ambient surroundings

and to shield the operators from possible exposure to fine powders and the laser beam.

The laser power used varies greatly, from a few hundred watts to 20 kW or more, depending on the particular material, feed rate, and other parameters.

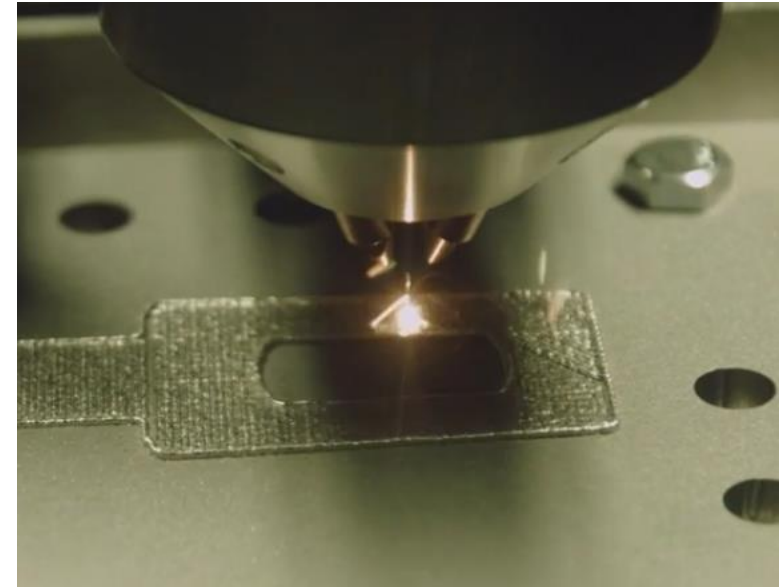


When a layer is completed, the deposition head moves up and continues with the next layer.

The process is repeated layer by layer until the part is completed.

The entire process is usually enclosed to isolate the process from the atmosphere.

Generally the prototypes need additional finishing, but are fully dense products with good grain formation.



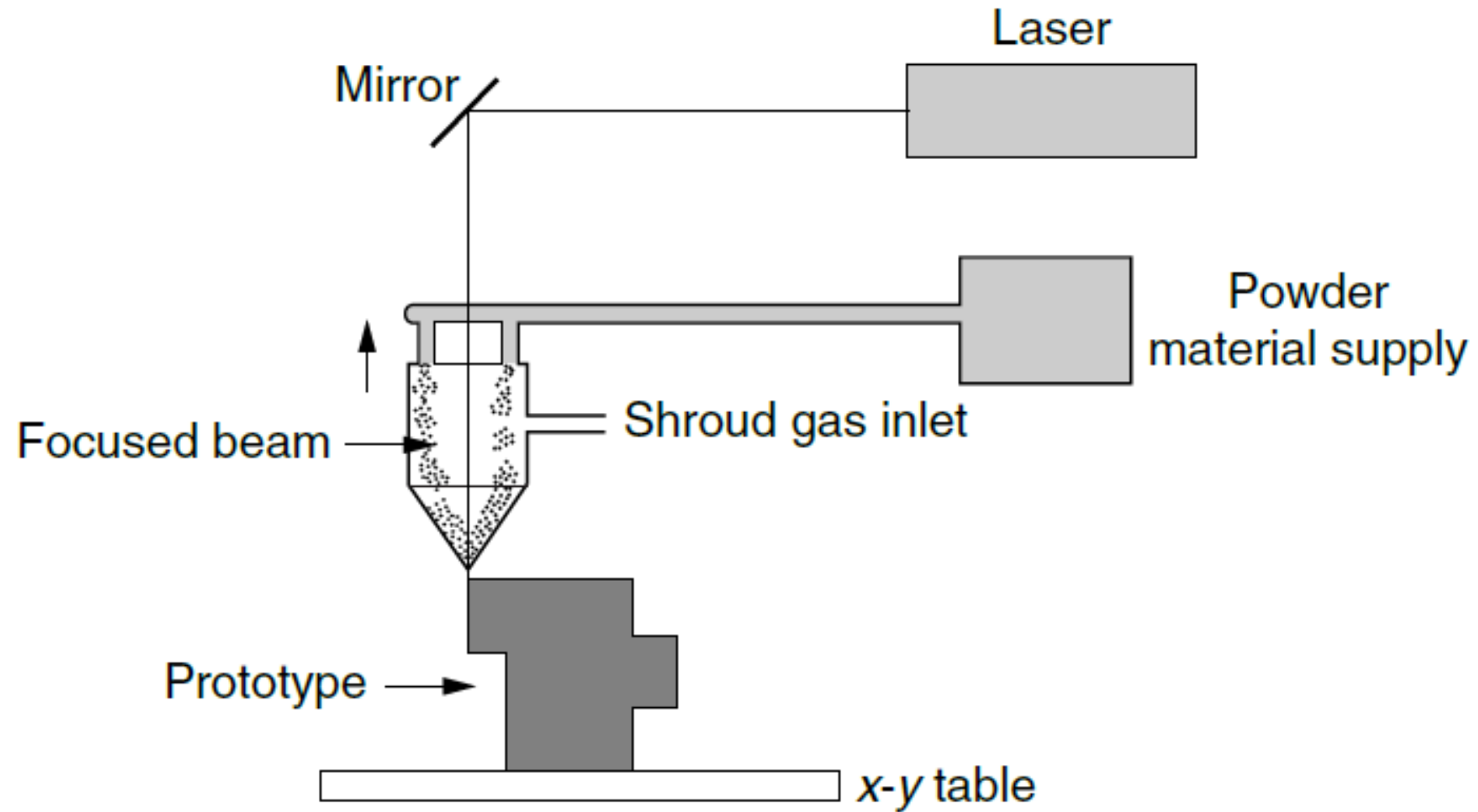
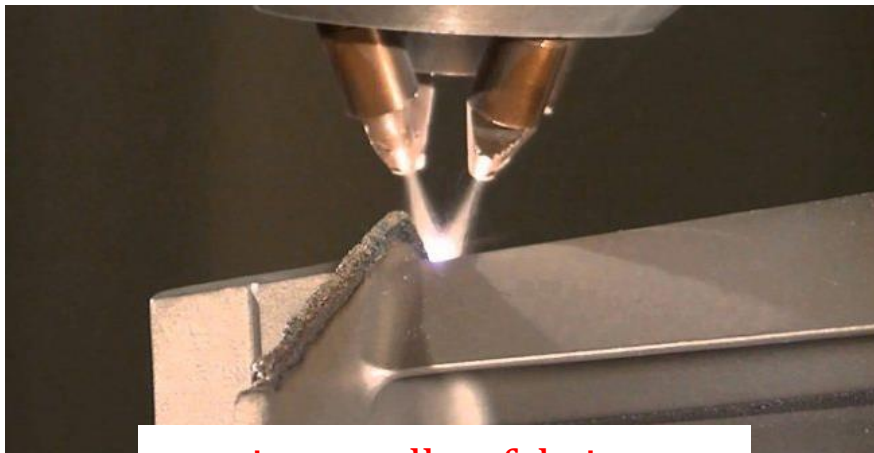


Figure 8.11 LENS process.

Objects fabricated are near net shape, but generally will **require finish machining**.

They are fully dense with **good grain structure** and have properties similar to, or even **better** than, the intrinsic materials

Laser engineered net shaping (LENS) has **fewer** material **limitations** than **SLS**, doesn't require secondary firing operations as some of those processes do, and can also be used to repair parts as well as fabricate them.



repair as well as fabricate



fabricated Objects

meaning
intrinsic
-belonging naturally

Material

A variety of materials can be used such as stainless steel, inconel, Copper, and aluminum.

Of particular interest are reactive materials Such as titanium.

Most systems use powder materials, but there has also been work done with material provided as fine wires.

In this case the Material is fed off-axis to the laser beam.



Advantages

(1) Superior material properties.

The LENS process is capable of producing **fully dense metal parts**. Metal parts produced can also include embedded structures and superior material properties. The microstructure produced is also relatively good.

(2) Complex parts.

Functional metal parts with complex features are the forte of the LENS system.

(3) Reduced post-processing requirements.

Post-processing is minimized, thus reducing cycle time.

Disadvantages

(1) Limited materials.

The process is currently narrowly focused to produce **only metal** parts.

(2) Large physical unit size.

The unit requires a relatively large area to house.

(3) High power consumption.

The laser system requires very high wattage.



Applications

Initial **applications** are concentrated on the fabrication

and repair of injection molding tools

and the fabrication of large titanium and other exotic metal parts for aerospace applications.



Applications

- Build mold and die inserts
- Producing titanium parts in racing industry
- Fabricate titanium components for biological implants
- Produce functionally gradient structures



Medical LENS produced metal part